

Ahmed Mahmoud Abdelaal

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SUMMARY

Passionate Software Engineer with strong expertise in full-stack web development, desktop applications, and computer vision. Skilled in JavaScript, Python, React, Node.js, and various development frameworks. Excellent problem-solving and collaboration abilities. Currently pursuing a degree in System and Biomedical Engineering while building innovative software solutions.

EDUCATION

Cairo University - Faculty of Engineering

Bachelor's degree in System and Biomedical Engineering

Cairo, Egypt

2021 – 2026

TECHNICAL SKILLS

Web Development: REST API, JavaScript, HTML5, CSS, Node.js, Express.js, React.js, MongoDB, PostgreSQL

Desktop Development: Python, PySide6, PyQt5, C++, OOP

Libraries & Tools: Jest, TailwindCSS, Redux, Material UI, Docker, Git, Bash Script

CERTIFICATIONS

- Back End Development and APIs [freeCodeCamp](#)
- JavaScript Algorithms and Data Structures [freeCodeCamp](#)
- SQL [HackerRank](#)
- REST API [HackerRank](#)

PROJECTS

Learnova - Educational Platform

Full-Stack Web Development

In Progress

React.js, Node.js, MongoDB

- Developed a scalable web application to facilitate online learning
- Full-featured LMS with course creation, module organization, lessons, quizzes, and assignments
- Built-in discussion forums, real-time progress monitoring, and achievement systems

Sociopedia - Social Media Platform

Full-Stack Development

Feb 2024

Node.js, Express.js, React.js, Material-UI, MongoDB

- Built dynamic full-stack social media platform
- Implemented JWT-based authentication for secure user access
- Responsive design with modern UI components

ChatEase - Real-Time Chat Application

Full-Stack Development

Aug 2024

React, Node.js, Socket.io, MongoDB

- Full-featured messaging platform with real-time capabilities
- Implemented JWT-based authentication and real-time messaging
- Added media upload functionality for enhanced user experience

Redis-monitor-dashboard

Jun 2025

Real-time Redis Monitoring Dashboard

JavaScript, React, WebSockets

- Real-time monitoring of Redis metrics with live WebSocket updates
- Interactive charts for memory usage, operations per second, and connected clients
- User-friendly dashboard for tracking Redis performance at a glance
- Live updates and visualization for effective troubleshooting and analysis

LucidumLab - Computer Vision Initiative

May 2025

Open-Source Computer Vision

Python, OpenCV, NumPy

- Open-source initiative focused on computer vision projects and algorithms
- Implemented image filtering techniques including Gaussian, median, and bilateral filters
- Developed edge detection algorithms using Sobel, Canny, and Laplacian operators
- Created face recognition system with feature extraction and matching capabilities

FilterLab - Digital Filter Designer

Jan 2025

Signal Processing Application

Python, PyQt5, Matplotlib, SciPy

- Built interactive desktop application for digital filter design and analysis
- Supports design of standard filters (low-pass, high-pass, band-pass) and all-pass filters
- Real-time visualization of filter responses, pole-zero plots, and frequency characteristics
- Generates corresponding C code implementation for designed filters

TuneSpy - Audio Analysis Tool

Jan 2025

Audio Processing Application

Python, NumPy, Matplotlib, LibROSA

- Developed comprehensive audio analysis and fingerprinting application
- Generates detailed spectrograms and extracts MFCC (Mel-Frequency Cepstral Coefficients) features
- Implements audio fingerprinting algorithm for similarity matching
- Compares loaded audio files with preprocessed database to find similar matches

EXTRACURRICULAR AND ACTIVITIES

Software Engineer

Jan 2025 – Present

IEEE EMBS Student Branch - Cairo University

Cairo, Egypt

- Mentored students in backend development technologies
- Conducted workshops and hands-on sessions on Node.js, MongoDB, Express.js
- Led student activities and technical training programs

Fullstack Developer

NASA SpaceApps Hackathon

Exosky Project Team

Remote

- Developed Exosky web application exploring exoplanets and stars using NASA APIs
- Implemented 3D visualizations and search functionality
- Created interactive drawing constellations mode for educational purposes

LANGUAGES

- Arabic: Native/Fluent
- English: CEFR B2 – Upper Intermediate